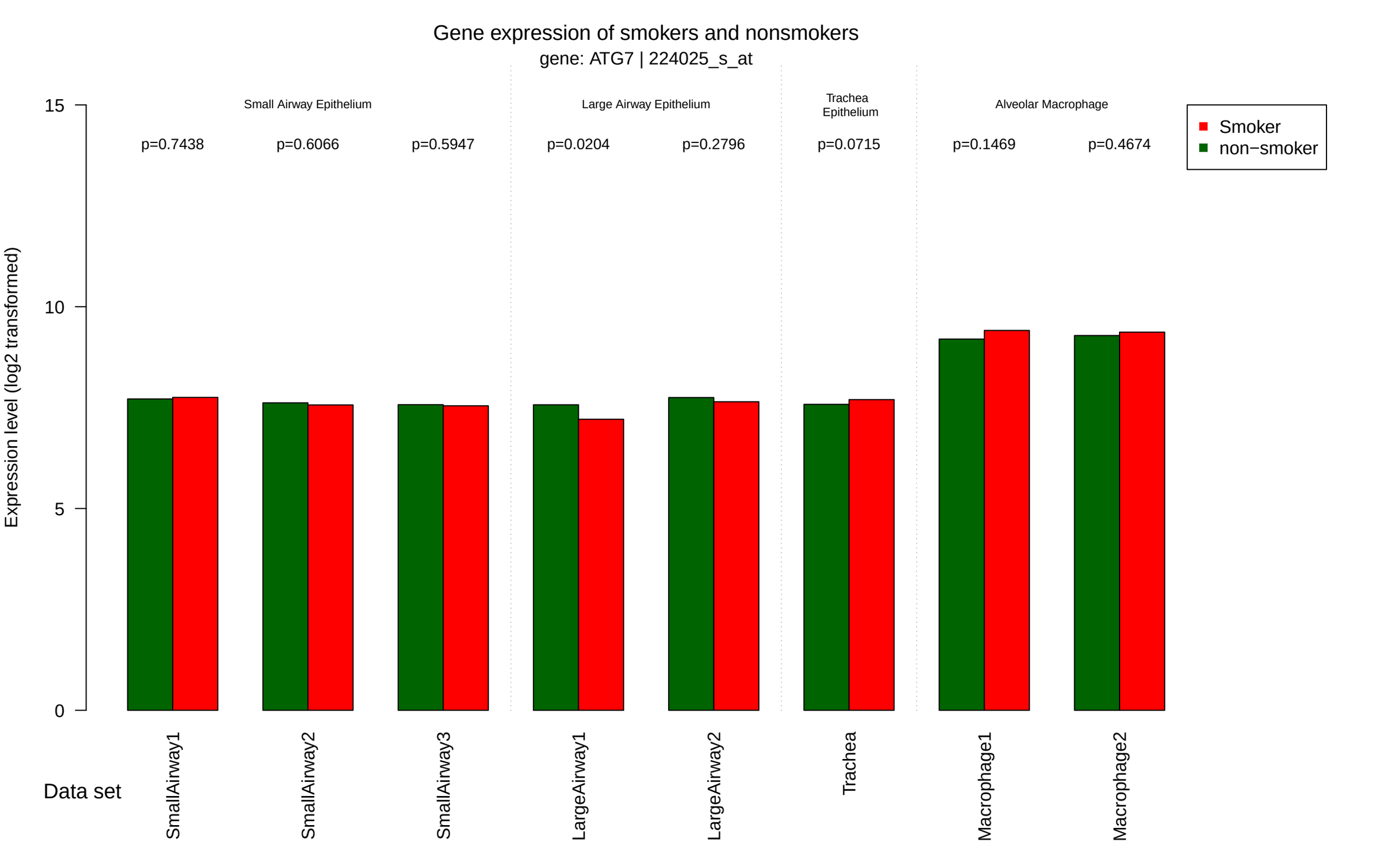
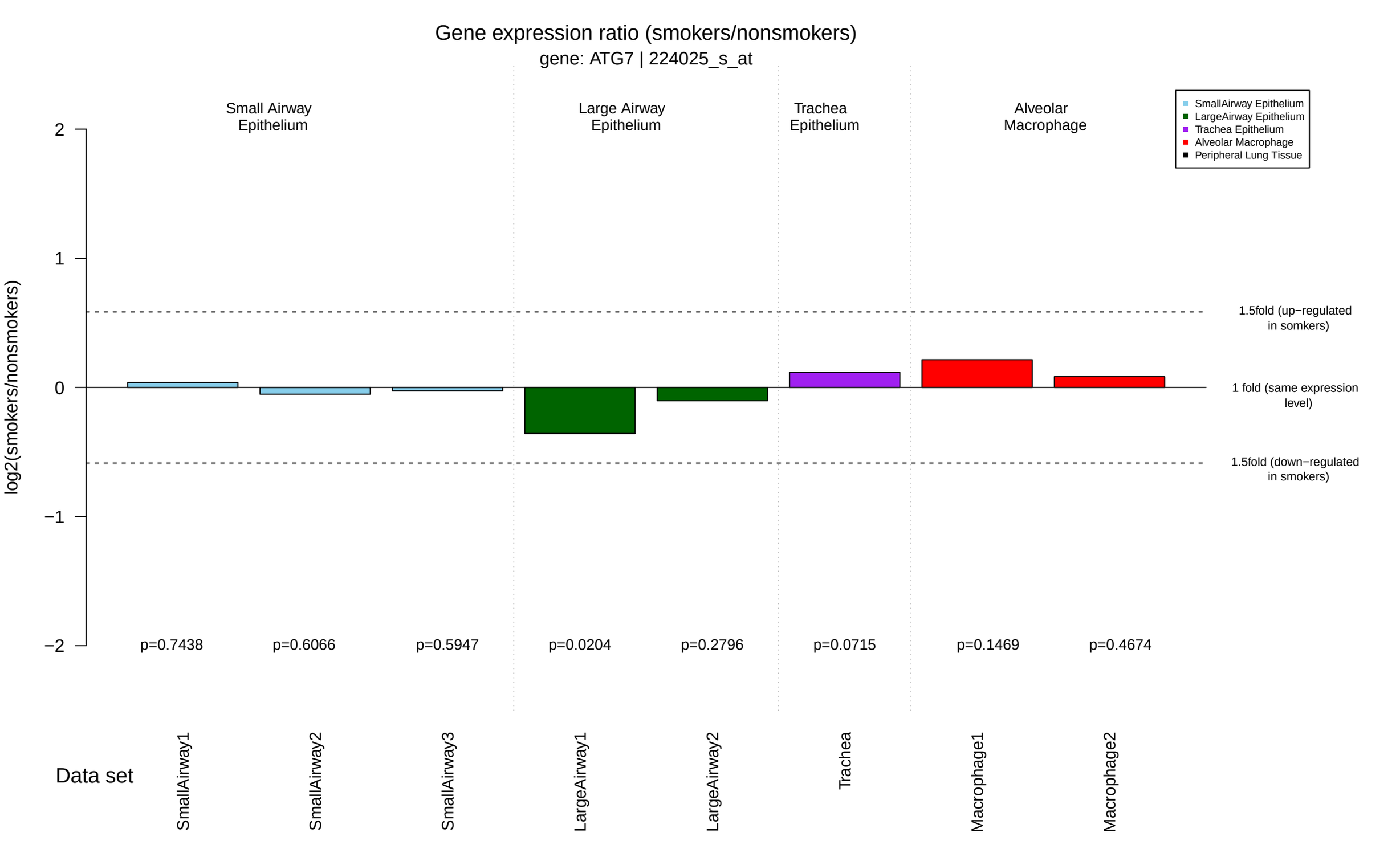
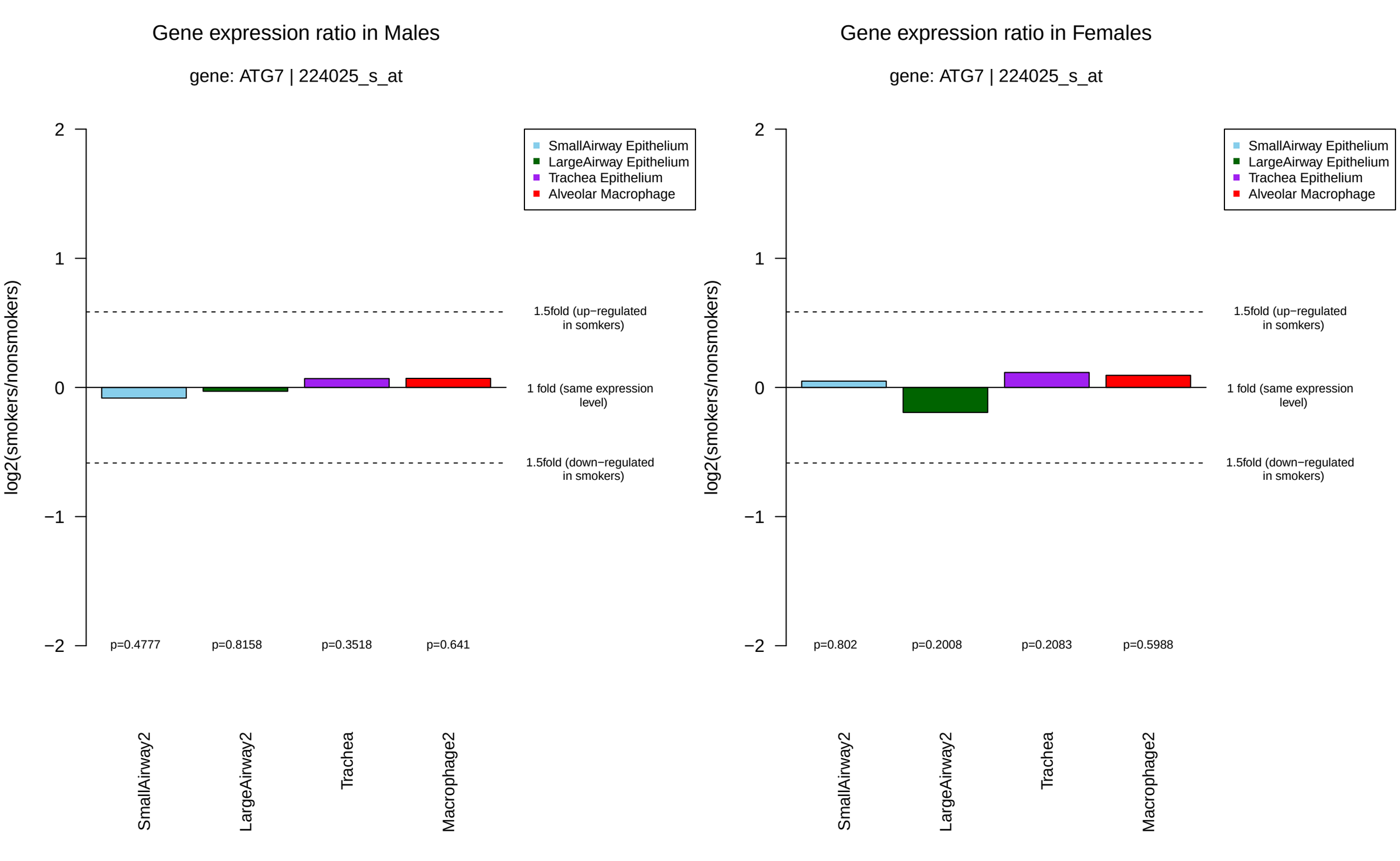
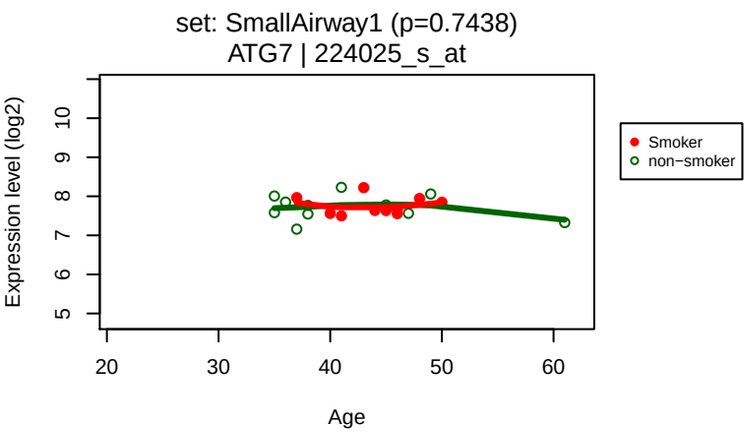
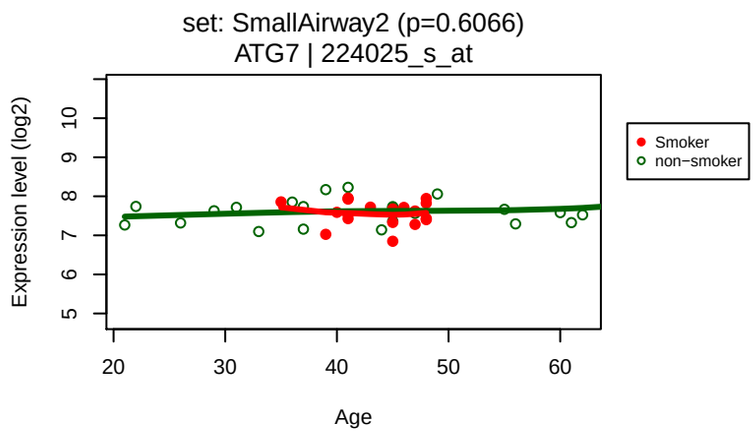
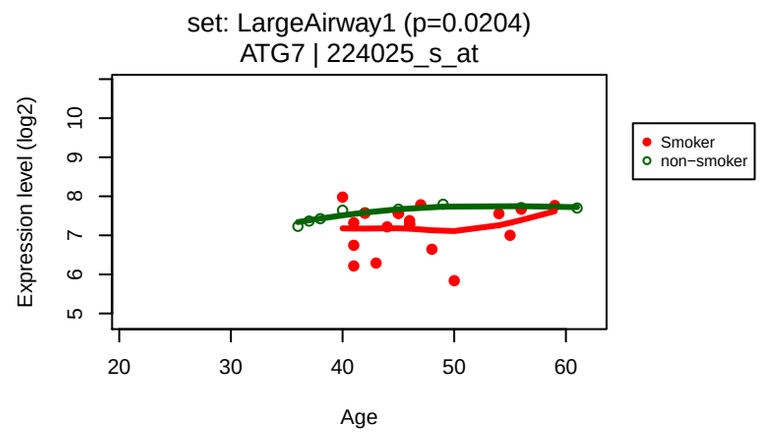

Gene expression of smokers and nonsmokers
gene: ATG7 | 224025_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7438
p=0.6066
p=0.5947
p=0.0204
p=0.2796
p=0.0715
p=0.1469
p=0.4674
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

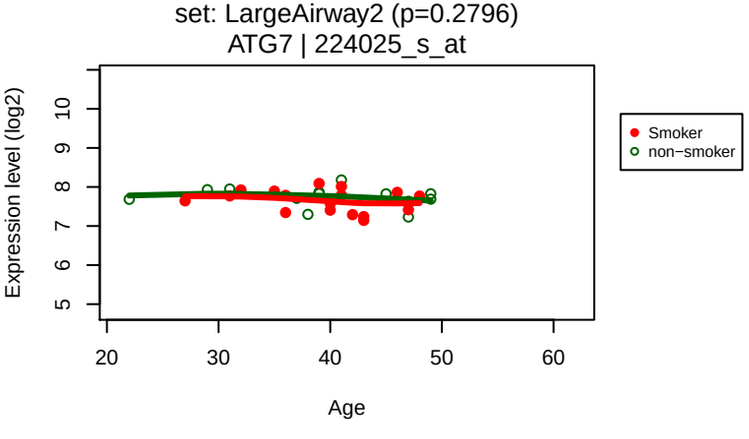

Gene expression ratio (smokers/nonsmokers)
gene: ATG7 | 224025_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7438
p=0.6066
p=0.5947
p=0.0204
p=0.2796
p=0.0715
p=0.1469
p=0.4674
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

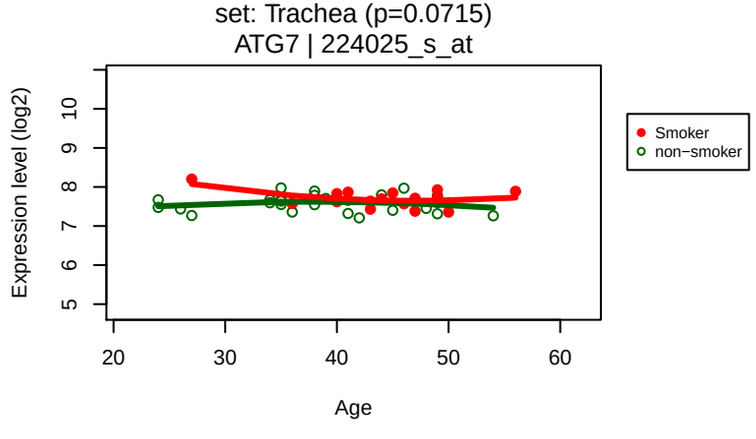

Gene expression ratio in Males
gene: ATG7 | 224025_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4777
p=0.8158
p=0.3518
p=0.641
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATG7 | 224025_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.802
p=0.2008
p=0.2083
p=0.5988
SmallAirway2
LargeAirway2
Trachea
Macrophage2

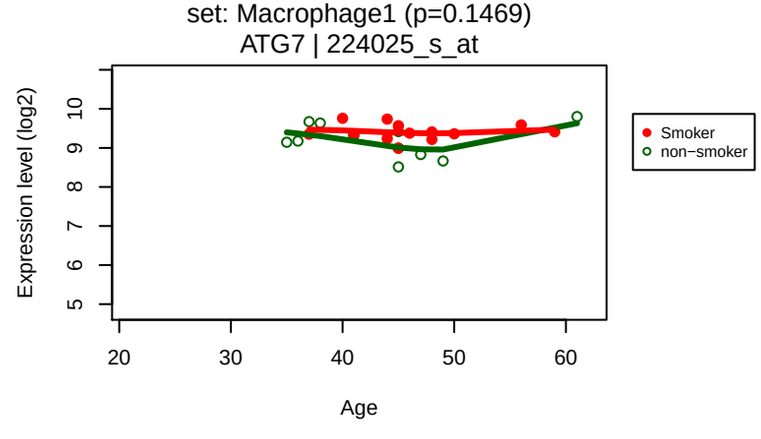

set: SmallAirway1 (p=0.7438)
ATG7 | 224025_s_at
Expression level (log2)
Age
Smoker
non−smoker

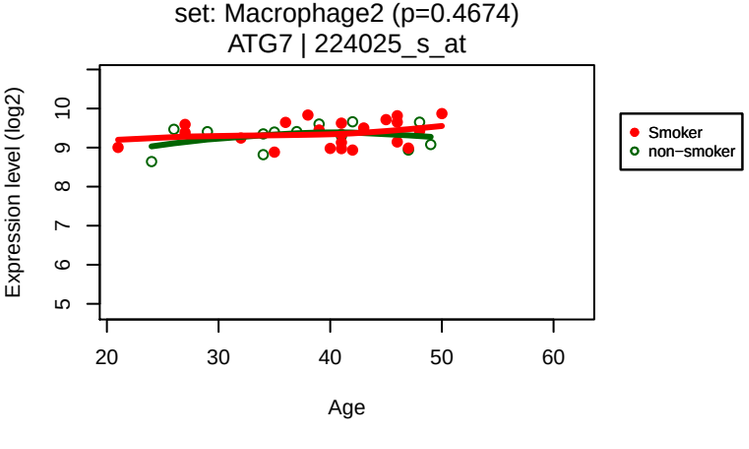

set: SmallAirway2 (p=0.6066)
ATG7 | 224025_s_at
Expression level (log2)
Age
Smoker
non−smoker

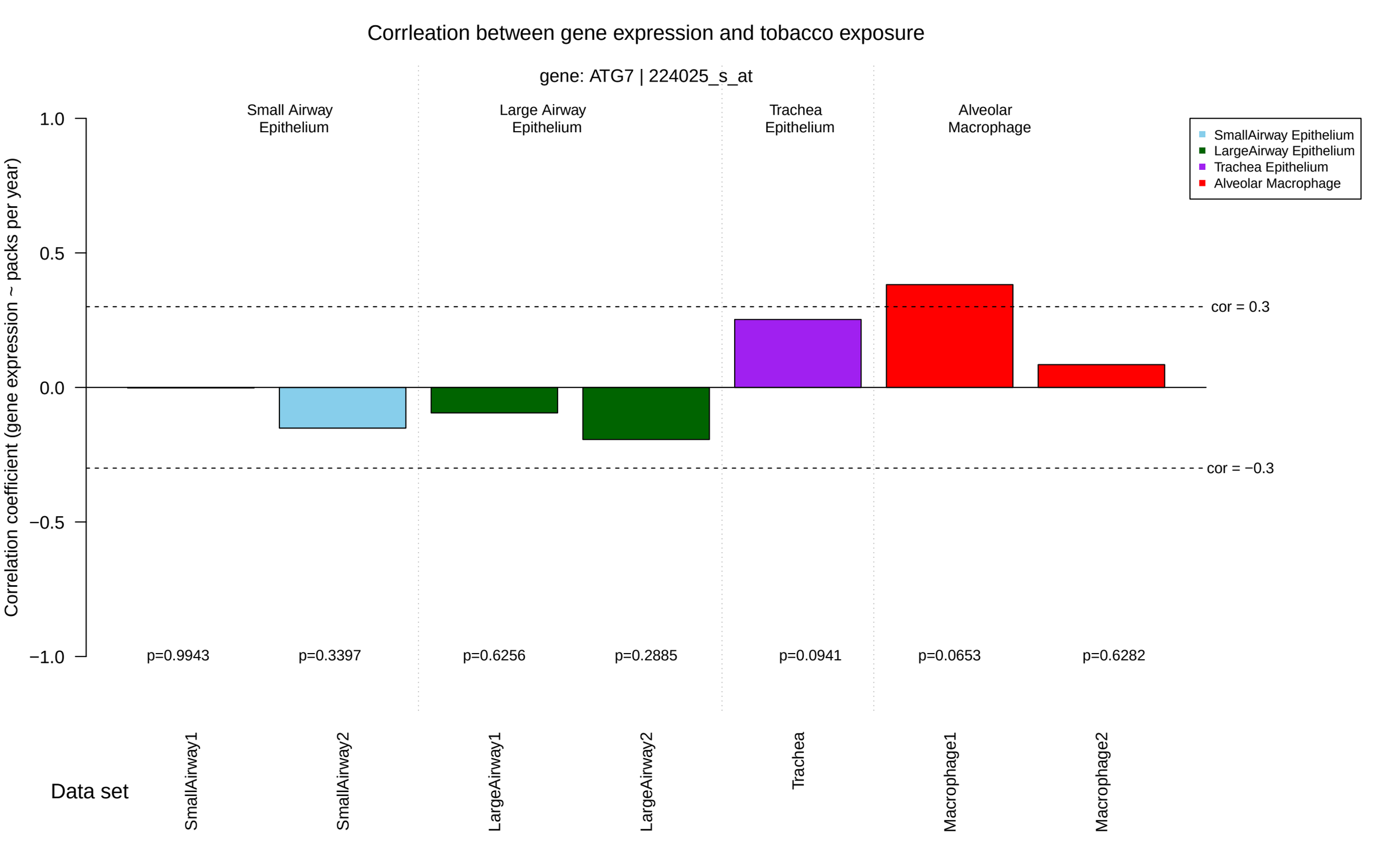

set: LargeAirway1 (p=0.0204)
ATG7 | 224025_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2796)
ATG7 | 224025_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0715)
ATG7 | 224025_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1469)
ATG7 | 224025_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4674)
ATG7 | 224025_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATG7 | 224025_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9943
p=0.3397
p=0.6256
p=0.2885
p=0.0941
p=0.0653
p=0.6282
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2